# Half of Your CMDB's Measured Value Is Knowing Who Opened the Ticket

**Sudhir Dixit**
Independent Researcher
sudhir.dixit1@gmail.com

**Abstract**

Configuration management database (CMDB) programmes are justified by the analytics they enable, and that justification is a comparison against a baseline the analyst chooses. On a public event log of $45{,}455$ incidents from a bank's IT operation whose affected-item field is $100\%$ populated, we show the choice is worth nearly as much as the data itself. Predicting at intake whether an incident will be reassigned, knowing the affected configuration item is worth $+0.183$ AUC against four intake fields, and $+0.103$ $[+0.094, +0.113]$ once one more free field is added: which group logged the incident. The gap survives three estimator families, including one in which item identity is a single column rather than $2{,}554$, and reproduces on two further targets and at three target thresholds. Stated as a detection problem it is larger than the AUC ratio of $1.8$ suggests — at a review capacity of $5\%$ of arrivals, omitting the field credits the CMDB with $3.9$ $[3.2, 6.4]$ times as many additional catches. Two measurements explain the overlap: the opening group adds under $0.01$ AUC to a model that already knows the item, and the shrinkage is graded in the field's resolution, running $27\%$ to $44\%$ as it goes from one bit to $49$ levels. We also report three things we got wrong: a null that could not fail, an operational factor we first reported without an interval, and a field whose meaning we mis-stated in an earlier draft.

## 1 Introduction

A CMDB programme is expensive and is justified by what it will enable. That justification rests on a comparison whose second arm — performance without configuration data — is a choice the analyst makes, usually silently, by deciding which already-available fields to include.

On one organisation's incident log the choice halves the answer, and we can say exactly where the difference goes. This is a short paper with one result and one mechanism.

**What kind of contribution this is.** We report no deployed system. This is a retrospective study on a public benchmark, offered as an evaluation practice: the quantity we measure is not a model's accuracy but the number a deployment decision turns on, and that number is set by a field-admission choice rarely written down. Section 6 states what it costs in the units a desk operates in.



**On what this paper used to claim.** Earlier versions reported larger findings withdrawn under adversarial review, each an artifact of a control we built ourselves. Three corrections survive here as reported results rather than silent edits, because each is the kind of error this paper is about: a null that could not fail (Section 5), a factor reported without an interval (Section 6), and a central field whose meaning we asserted rather than checked (Section 3).

## 2 Background

**The CMDB and what justifies it.** A CMDB records configuration items and their relationships, and is the asset on which the configuration-management practice of ITIL and its successors rests. Survey work on ITSM framework adoption (Marrone and Kolbe 2011) finds that realised benefits grow with implementation maturity, but measures them as practitioner-reported outcomes rather than as the analytic value of the configuration data itself. We are not aware of a peer-reviewed measurement of what knowing the affected configuration item is worth to a named prediction task; the material addressing that question directly is vendor documentation. The gap is the occasion for this paper, and part of the reason it exists is that the data to close it barely does: of the three public ITSM incident logs in common use, one records the affected item on every incident, the second records it on $0.2\%$, and the third has no such field at all (van Dongen 2014; Amaral, Fantinato, and Peres 2018; Steeman 2013). That is also why this study is single-organisation, and it is a constraint rather than a choice.

**The methods a CMDB is bought to enable.** Surveys of AIOps for failure management (Zhang et al. 2024; Remil et al. 2024; Notaro, Cardoso, and Gerndt 2021) catalogue triage, failure-prediction and root-cause methods that take configuration data as input. Incident triage in particular — deciding which team owns an incident — has been treated as a learning problem on operational logs at industrial scale (Chen et al. 2019), as has failure prediction across a cloud estate (Lin et al. 2018).

**Prediction on incident event logs.** Our task sits in predictive process monitoring, surveyed and benchmarked across real-world logs (Teinemaa et al. 2019; Verenich et al. 2019) on data of the kind process mining formalises (van der Aalst

2016). The BPI Challenge logs are the standard public instances (van Dongen 2014; Steeman 2013); comparable exports exist elsewhere (Amaral, Fantinato, and Peres 2018). Such logs carry documented quality hazards (Suriadi et al. 2017), including values denormalised from a record onto its events, which we rule out below.

**What the comparison is measured against.** The value of a feature *given* a baseline is what algorithm-agnostic variable-importance frameworks formalise (Williamson et al. 2021), and our overlap is the interaction additive importance measures handle (Covert, Lundberg, and Lee 2020). That measured importance depends on what the model already has is therefore not news; what is not established is its magnitude for configuration data on a named operational task, or that the field it overlaps with is one the organisation already records for nothing. Rendle, Zhang, and Koren (2019) and Ferrari Dacrema, Cremonesi, and Jannach (2019) show inadequately specified baselines accounting for reported gains in recommender systems; benchmark conclusions are sensitive to variation that is seldom reported (Bouthillier et al. 2021), and documentation of the choices producing a result is thin (Gundersen and Kjensmo 2018). Where the choice concerns which fields may enter, it shades into leakage (Kaufman et al. 2012; Kapoor and Narayanan 2023), which for predictive process monitoring needs explicit construction to avoid (Weytjens and De Weerdt 2022). We report this failure mode for a deployment decision rather than a leaderboard, where field admission is not a modelling detail but the number the business case turns on.

## 3 Data and Task

We use the incident records of the BPI Challenge 2014 log (van Dongen 2014) from Rabobank Group ICT, recorded in HP Service Manager, with the activity log shipped alongside.

**Cleaning.** Of 46,809 rows, 203 carry no incident identifier and one an unparseable timestamp. A further 1,150 opened before 2013-10-01 are left-censored — already open when the extract began — and show reassignment rates of 76–100% against a subsequent 35–42%. Removing them leaves **45,455** incidents to 2014-03-31; a temporal 70/30 split gives 31,818 training and 13,637 test, positive rates 0.413 and 0.372. The cutoff is not taken from the outcome: monthly volume rises from 857 incidents in September 2013 to 8,606 in October. The affected-item field is 100% populated across 2,929 items, 2,554 seen in training.

**Task and estimator.** Predict at creation whether an incident will later be reassigned. We call the target reassignment rather than misrouting throughout: it fires on 37.2% of test incidents, which is routine handling and not error, and Section 6 reports the ladder at stricter thresholds. Median handling time rises from 1.59 hours for incidents never reassigned to 47.53 hours for those reassigned five or more times; we use this to motivate the task, not as a recoverable saving. The primary estimator is one-hot logistic regression, chosen because it is bin-free; two further estimator families are reported in Section 4. Encoders and rankings are fit on training data only; intervals are 2,000-resample paired bootstraps, except Table 2, whose counts are re-ranked in each draw and use 400.

| Baseline | AUC | + item id. | gain [95% CI] |
|---|---|---|---|
| intake only | 0.562 | 0.746 | +0.183 [+0.172, +0.195] |
| + opening group | 0.644 | 0.748 | +0.103 [+0.094, +0.113] |

Table 1: The value of knowing the affected configuration item, with and without a field the service desk already records.

**The free field, and what it is not.** The activity log records, on the `Open` event of every incident, an assignment group: 49 in training, none missing. An earlier draft of this paper called it the queue the service desk routed the incident to. That was an assertion, not a finding, and it is wrong. Three checks say so. The `Open` rows carry 50 distinct groups where the `Assignment` rows carry 218, and a routing destination cannot be less diverse than the teams that then do the work. One group accounts for 67.0% of `Open` rows but only 18.4% of all activity rows. And the `Open` group matches the group on the incident's first `Assignment` activity for just 15.1% of the 37,577 incidents that have one. The field is the group that *logged* the incident — a central service desk for two thirds of tickets, a specialist team opening its own work for the rest. It is also concentrated and unstable: the largest group holds 62.1% of training incidents, and the number of live groups falls from 49 to 32 between our two halves.

We report the error because the correction carries the paper's method. Nothing measured changes; what changed is a claim about what the field *meant*, made on no evidence, in the way this paper accuses business cases of treating baselines.

**Why not the real routing decision.** The obvious repair — use the first `Assignment` group — is not available: it occurs strictly after `Open` on every incident that has one, a median of 46 minutes later, and 7,878 of 45,455 never have one. It fails the criterion below, and substituting it would introduce the leakage that criterion prevents.

**Admissibility, and mutation.** A field on the `Open` row is admitted only if it is a per-event observation, tested by whether it varies across an incident's activity rows; the opening group varies for 92.56% of cohort incidents and matches the last-observed group for only 21.35%. Because the incident file is a closed-record snapshot we also report how often each field is edited after creation: Urgency on 1,107 incidents (2.44%), Impact on 1,084 (2.38%), the affected item on 159 (0.35%). Incidents whose Impact was edited are reassigned at 0.66 and those whose Urgency was edited at 0.67, against 0.39 for the rest, so this is not neutral; Section 4 reports the headline with those fields dropped and with the edited incidents removed.

## 4 The Result

Against four intake fields the affected item looks decisive. Adding the opening group cuts its measured value by 44%.

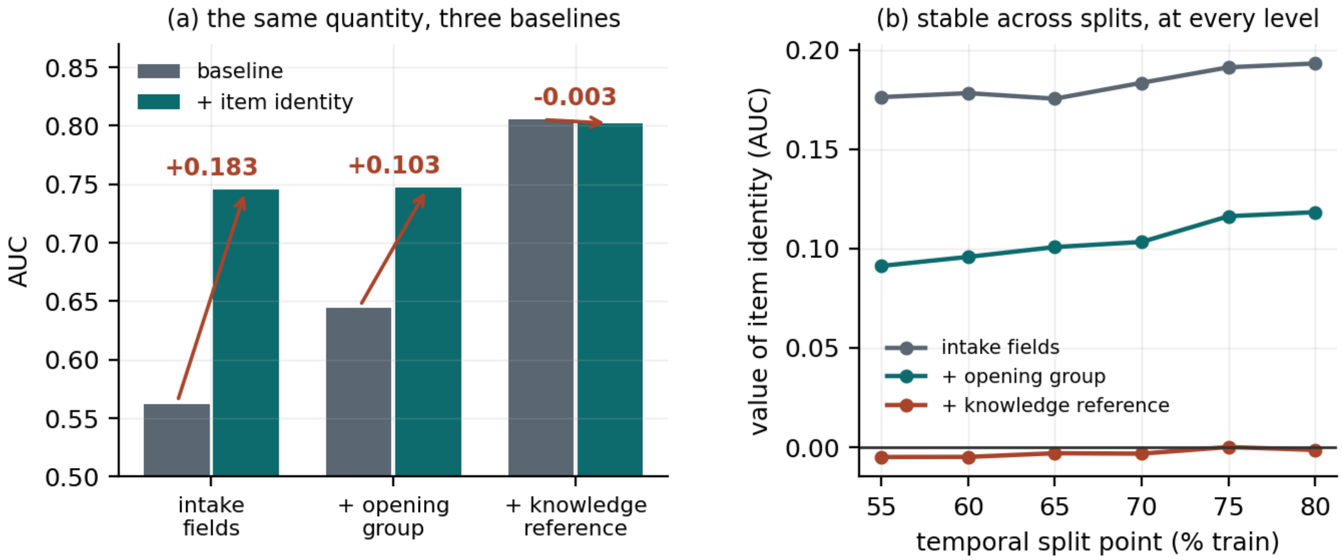


Figure 1: (a) The same quantity under three baselines; the third is discussed in Section 8. (b) The ordering holds at every temporal split point from $55\%$ to $80\%$.

The gains are roughly 28 and 17 standard deviations outside a null in which the incident-to-item association is shuffled, preserving column count and marginal distribution while destroying the signal — we round because re-drawing the same null at a different draw count moves the first figure by about $0.7$ — a comparison that matters because adding item identity adds $2{,}554$ indicator columns, which is not free under a fixed penalty. These $z$ figures pool the null's Monte-Carlo dispersion with the gain's own bootstrap standard error; dividing by the null alone, as an earlier draft did, would report 51 and 33.

It is not a split artifact: across six split points the two gains range $+0.175$ to $+0.193$ and $+0.091$ to $+0.118$ (Figure 1b), and across split point, target threshold and cleaning cutoff the second ranges $+0.068$ to $+0.130$ — wider than any bootstrap interval, which conditions on all of those. Nor does the disclosed field editing explain it: reducing the intake block to Category alone gives $+0.106$, and restricting to the $44{,}227$ never-edited incidents gives $+0.107$.

**It is not an artifact of the estimator.** The dimensionality null rules out the regularisation burden of $2{,}554$ sparse columns, but a reader may want the effect shown where that burden cannot arise. We re-represent item identity as a single cross-fitted target-encoded column — fitted on training only, out-of-fold, so no row sees its own outcome — which makes the item one continuous feature and makes histogram boosting usable on a variable that otherwise collapses $2{,}554$ items into 137 bins. Across one-hot logistic regression, logistic regression on the encoded item, and boosting on it, the first rung ranges $+0.173$ to $+0.183$, the second $+0.092$ to $+0.103$, and the shrinkage $43\%$ to $47\%$. Encoding a *shuffled* item column under the same pipeline returns $-0.0001 \pm 0.0016$ and $+0.0042 \pm 0.0020$. The first collapses to zero; the second does not quite, and we report it rather than round it away

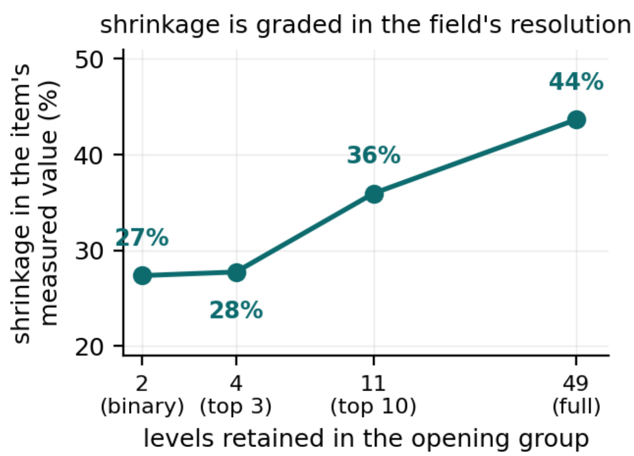


Figure 2: Coarsening the opening group and re-running the ladder. The shrinkage in the item's measured value is graded in the field's resolution, which is what a proxy relationship predicts and a coincidence does not.

because it bounds how much of the boosting rung could be encoding artifact.

## 5 Where the Difference Goes

The two rows differ by $0.080$, which is not itself informative: for any two variables, the drop in one's gain when the other joins the baseline equals the drop in the other's. Read four ways, the item is worth $+0.183$ over intake and $+0.103$ once the group is present; the group is worth $+0.082$ over intake and $+0.002$ once the item is present. Two measurements explain the overlap.

**First**: adding the opening group to a model that already knows the item is worth $+0.0017$ $[+0.0001, +0.0034]$. It clears a matched-dimension null of $-0.0009 \pm 0.0005$ that no draw reaches, so it is not the cost of adding columns; but across split points and penalties it ranges $+0.0002$ to $+0.0072$, so the honest statement is that its unique contribution is *under* $0.01$ *AUC and not resolvable more finely*.

**Second**: the group's predictive content is largely a function of the item. Randomising the group label *within* each item — which destroys the label's own identity while preserving what the item implies about it — retains $91\%$ of the group's $+0.082$ gain.

**A floor that could not fail.** An earlier draft compared that $91\%$ against a floor of $2\%$ and claimed a margin of $89$ points. The floor was wrong. It drew cell labels independently *per row*, so two incidents sharing an item landed in different cells and the association was destroyed by construction: the null could only ever return zero. Rebuilt as a random partition *of items*, which preserves item-level grouping and knows nothing about who opened anything, the floor rises with the partition's fineness — $41\%$ at $49$ cells, matching the field's own cardinality, $77\%$ at $400$ (Figure 3b). It is a 30-draw Monte Carlo and its dispersion is wide — $12$ points at $49$ cells — so the margin below is indicative, not resolved. Note what this sweep cannot show: at one cell per item the null *is* the real leg, so "the real leg beats every floor" would be true by construction. The informative comparison is at matched cardinality, where a grouping that knows nothing about intake retains $41\%$ against the real $91\%$ — a margin of $50$ points, not the $89$ we published against a null that could only return zero.

**How far this goes, and where it stops.** Without any estimator, item identity carries $60.4\%$ of the opening group's information and the group carries $19.6\%$ of the item's. Both are plug-in estimates over thousands of conditioning levels and inflated by finite samples — shuffling item identity leaves floors of $14.0\%$ and $4.5\%$ — so read the asymmetry, which survives with $46$ points against $15$, not the levels. Nor is it determination: a modal-group lookup on item identity reproduces which group logged the incident on $90.3\%$ of test incidents against $78.6\%$ for always guessing the largest, and class-balanced reaches only $34.1\%$. Item identity resolves the coarse contrast — central desk or specialist team — and much less of the fine one.

**The shrinkage is graded in the field's resolution.** That reading makes a prediction, and it holds. Coarsening the group and re-running the ladder gives shrinkages of $27\%$, $28\%$, $36\%$ and $44\%$ at two, four, eleven and $49$ levels (Figure 2). A single binary flag — central desk or not — already recovers $61\%$ of the group's baseline gain and $63\%$ of the shrinkage. The field that must be admitted is close to one bit, far cheaper to reproduce elsewhere than a 49-way taxonomy.

**Is the field really free?** One objection goes to the heart of the reading, and our own evidence sharpens it. The opening group may be free only because a human at the desk already knew what the ticket was about — in which case it is undocumented configuration knowledge, and a baseline containing it is not a CMDB-free baseline. The one-bit result makes this concrete: the contrast doing most of the work is central desk against a specialist team opening its own work, exactly where such knowledge would sit. We cannot settle it and do not need to. The measurement is the same under both readings; what changes is the moral. If the group is independent, a CMDB is worth half what the naive comparison says. If it is tacit configuration knowledge, then our $+0.103$ is a lower bound and the organisation already holds much of what the CMDB would tell it, in a form it is not paying to maintain. We do not choose between these: the log cannot identify which is true, and the reading changes what the number means. What survives either way, and is the claim we make, is that the business case cannot be written against the intake block alone.

| Capacity | honest extra | naive extra | factor |
|---|---|---|---|
| 5% | $+67$ $[43, 84]$ | $+262$ $[242, 300]$ | 3.9 $[3.2, 6.4]$ |
| 10% | $+33$ $[6, 70]$ | $+353$ $[304, 404]$ | 10.7 $[5.5, 39.9]$ |
| 20% | $+18$ $[-28, 82]$ | $+481$ $[407, 524]$ | 26.7 $[5.8, 221.0]$ |

Table 2: Extra reassignment-bound incidents surfaced by adding item identity, against a baseline that includes the opening group (honest) and one that omits it (naive). Paired bootstrap over test rows, with ties at the capacity cut broken at random in every draw. The point estimates are one such draw; the intervals average over 400.

**What we do not claim.** The reverse framing — that the item column "stands in for" the group — is not established. We measured it, by randomising items within groups, and obtained $44\%$ of the item's gain — but a routing-blind equal-size partition of items retains *more*, $56\%$, and one whose cell sizes follow the group's own mass profile retains $25\%$. The leg is floor-dominated and we exclude it. (An earlier draft reported a negative value for the second floor. That construction drew cells per row, the same defect described above, and the equal-size floor was computed from different partitions for training and test; both are corrected here.) Nor do we claim a direction of causation: the log records no counterfactual, so whether the item drives who logs the ticket or both follow a third cause is not identified here. Our claim is about what an analyst's baseline choice does to a measured number, which does not depend on which is true.

**Two fields we did not admit, and why.** Our baseline is not a terminus either. Hour of day and day of week are free and creation-time; adding both moves the item's value from $+0.103$ to $+0.099$. `Service Component WBS (aff)` is harder: $100\%$ populated, free, and admitting it takes the measured value to $+0.023$. We exclude it as a near-deterministic function of the item — only $58$ of $2{,}929$ items carry more than one value — so it is configuration data under another name. A reader who disagrees should read our headline as $+0.023$, and that substitution is the decision this paper is about.

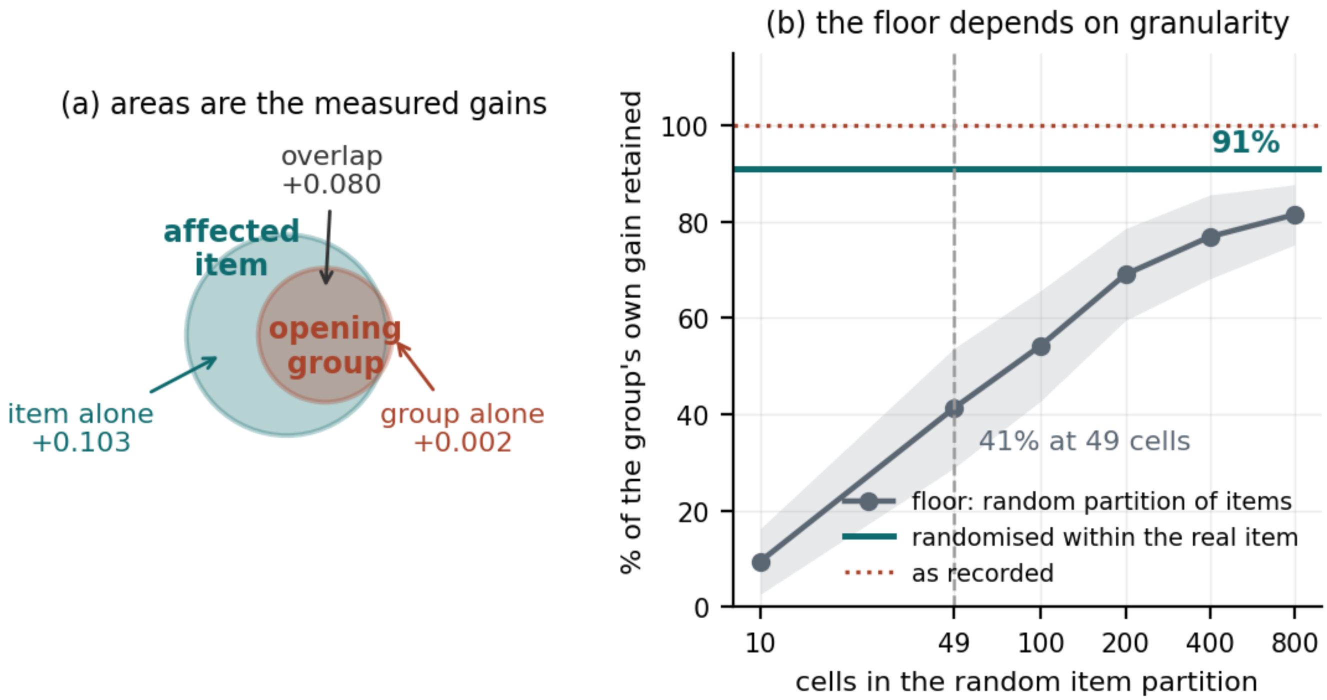


Figure 3: (a) The overlap, drawn to scale: each circle's area is that field's measured gain over the intake block, and the centre distance is solved so that the intersection equals the measured overlap. The opening group lies almost entirely inside the item. (b) Randomising the group label within each item costs little; the same randomisation within a random partition of items costs more, by an amount that depends on how fine the partition is.

## 6 What the Difference Buys

An AUC difference is not a quantity a service desk can act on, so we state the same comparison as detection. Fix a review capacity — the share of arrivals a desk can look at before routing — rank the test incidents by predicted risk, and count how many reassignment-bound incidents each model surfaces. Both arms use matched models: the honest contrast adds item identity to intake-plus-group, the naive one to intake alone, mirroring Table 1. An earlier draft did not match them, crediting item identity with the group's own contribution.

At $5\%$ capacity, knowing the affected item surfaces 67 $[43, 84]$ more reassignment-bound incidents than the intake-plus-group baseline does, or about 36 a month at this log's volume. An analyst who had omitted the group would have credited item identity with 262 $[242, 300]$. The omission overstates the operational gain by a factor of 3.9 $[3.2, 6.4]$, against a ratio of only $1.8$ between the same two gains measured as AUC.

The factor grows with capacity — $10.7$ at $10\%$, $26.7$ at $20\%$ — but so does its interval: the honest denominator shrinks to a few dozen incidents, at $20\%$ it includes zero, and both of those factor intervals are conditioned on a positive denominator, which the $5\%$ row does not need. We quote the $5\%$ figure and claim only that the overstatement is several-fold and larger than the AUC ratio implies. It is larger because AUC averages over every operating point while a capacity-limited desk works at one, near the top of the ranking where the group-aware baseline is already near its ceiling. None of this says a surfaced incident is a corrected one: the log records no intervention.

**Three target definitions.** Because reassignment is common, we repeat the ladder at stricter thresholds. Requiring two or more reassignments ($21.8\%$ of incidents) gives $+0.131$ and $+0.068$; three or more ($10.6\%$) gives $+0.151$ and $+0.080$. The magnitude is threshold-dependent and not monotone, but the interval on each +group rung excludes zero (we compute intervals for that rung only), the ordering never reverses, and the shrinkage stays between $43\%$ and $49\%$. The transferable quantity is the shrinkage, not the gain.

## 7 Does It Survive a Change of Task?

A gap on one target may be a property of that target, so we repeated the ladder on two further outcomes, changing only what is predicted. *Reopening* is close to an independent failure mode: it fires on 2,096 incidents and correlates with reassignment at $+0.14$. *Long handling* — handle time above the training third quartile — correlates at $+0.40$, so we report it as corroboration only.

The shrinkage reproduces on both. On reopening, item identity is worth $+0.083$ against the intake block and $+0.055$ once the group is admitted, a reduction of $33\%$; on long handling, $+0.118$ and $+0.078$, a reduction of $34\%$. Every rung sits outside its matched-dimension null, and across five split points the reduction ranges from $30\%$ to $39\%$ on these two targets and from $38\%$ to $47\%$ on the primary one. Reopening is rare and its gains stand only $5.5$ and $4.2$ pooled standard deviations from their nulls, so it establishes the effect's direction and not its size; and the reduction is largest on the target the opening group is most directly about, which is what Section 5 predicts. That is why we state the transferable claim as the gap's existence rather than its magnitude.

## 8 One Thing We Could Not Establish

A third field, the knowledge-article reference, sits on the same `Open` row, is $100\%$ populated, and costs nothing. Adding it to the baseline raises AUC to $0.805$ and drives the measured value of item identity to $-0.003$. Whether that figure belongs in Table 1 depends on whether the field is available at creation, and we could not determine this.

Two structural tests both fail. A field that varies within an incident is certainly a per-event observation; the knowledge reference never varies — but neither does `Interaction ID`, which has $45{,}426$ distinct values for $45{,}455$ incidents and is plainly a creation-time key, so constancy is evidence of granularity, not timing. Exact identity with the closed record fares no better: the knowledge reference matches its closed-record column for $100.000000\%$ of incidents, but so does `Interaction ID` for $99.997628\%$ of the $42{,}151$ single-interaction incidents — one mismatch against a pass mark of $100.000000\%$.

We therefore make no claim about this field. The $-0.003$ is also inside the dimensionality null of Section 4 and changes sign under penalty tuning, so it is not resolvable on either ground. This is the paper's thesis turned on itself, and that reading is correct: our headline is contingent on a field-admission decision we cannot adjudicate. The transferable part is the negative — a field being free, populated and present on an opening record does not make it creation-time. The collection ships an interaction detail file we did not obtain; if it records this field at creation the question is answerable, and our inability to settle it is a limit of the three files we hold, not of the export.

## 9 Estate Concentration

Required CMDB scope is bounded by a property of the estate that needs no model: the top $8$ items generate $30.2\%$ of incidents, the top $64$ generate $70.5\%$, and the top $128$ — $5.0\%$ of the training vocabulary — generate $82.0\%$, all computable from a frequency count before funding anything.

Identifying only those items recovers most of the measured value. Averaged over five temporal split points, the top $8$ recover $56\%$ of the $+0.103$, the top $64$ recover $88\%$ and the top $128$ recover $93\%$; across those splits the three range over $52$–$58$, $81$–$93$ and $90$–$96$ These are min–max spreads over a design choice, not bootstrap intervals, and we write them as ranges to keep the distinction. We report the mean and range rather than one split's curve because a single curve is not monotone at this resolution: at $k = 32$ the across-split spread is 9 points. The same holds with the group removed from the baseline — $89\%$ at $k = 64$ — so this is a property of the estate's concentration, not of our field-admission decision.

## 10 Limitations

One organisation — of necessity, not choice (Section 2) — one platform, one task, six months, on 2013 and 2014 data from an on-premises product whose estate, intake practice and tooling have all moved since. The values $+0.183$ and $+0.103$ are properties of this estate; the transferable claims are that the gap exists, that it is large, that it is graded in the free field's resolution, and that the mechanism runs from item to group — the group's predictive content is largely carried by the item. The reverse, that the item column stands in for the group, is the leg Section 5 excludes. Reassignment is a proxy for misrouting and also fires on legitimate escalation. The within-item shuffle preserves item membership but not other correlates of identity, so it bounds the proxy share rather than isolating it. The opening group's distribution drifts between our halves, so the group-aware baseline is being asked to generalise across a shift.

## 11 Conclusion

Knowing which configuration item an incident concerns is worth $+0.183$ or $+0.103$ AUC for predicting reassignment, depending on whether the baseline includes one free field: which group logged the ticket. Stated as detection at a fixed review capacity, the same choice moves the credited benefit several fold. The difference is an overlap: a model given the item has already been told most of what the group would say. A business case built on the first figure is not measuring a CMDB. It is measuring a CMDB plus a field the organisation was already recording for nothing.

## Use of AI Systems

An AI assistant (Anthropic's Claude) was used to implement and refactor analysis code, to construct adversarial challenges to intermediate claims — several withdrawals above originated that way — and to draft and edit prose the author reviewed and verified. All analyses were specified, run and checked by the author, who takes full responsibility for the content, including every numeric claim and reference.


## Acknowledgements

This work benefited from the review and domain challenge of a practitioner in IT service management; several analyses were withdrawn on objections raised there.


## Reproducibility

The dataset is public (van Dongen 2014). Code, figures and a checker that recomputes every quantity here — from a result file or from the raw data: https://github.com/sudhirdixit1/cmdb-routing-queue-baseline.

## References


Amaral, C.; Fantinato, M.; and Peres, S. M. 2018. Incident Management Process Enriched Event Log. UCI Machine Learning Repository. Extracted from the audit system of a ServiceNow instance; 141,712 events over 24,918 incidents. CC BY 4.0.

Bouthillier, X.; Delaunay, P.; Bronzi, M.; Trofimov, A.; Nichyporuk, B.; Szeto, J.; Mohammadi Sepahvand, N.; Raff, E.; Madan, K.; Voleti, V.; Ebrahimi Kahou, S.; Michalski, V.; Arbel, T.; Pal, C.; Varoquaux, G.; and Vincent, P. 2021. Accounting for Variance in Machine Learning Benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*.

Chen, J.; He, X.; Lin, Q.; Zhang, H.; Hao, D.; Gao, F.; Xu, Z.; Dang, Y.; and Zhang, D. 2019. Continuous Incident Triage for Large-Scale Online Service Systems. In *Proceedings of the 34th IEEE/ACM International Conference on Automated Software Engineering (ASE)*.

Covert, I.; Lundberg, S.; and Lee, S.-I. 2020. Understanding Global Feature Contributions With Additive Importance Measures. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*.

Ferrari Dacrema, M.; Cremonesi, P.; and Jannach, D. 2019. Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys)*.

Gundersen, O. E.; and Kjensmo, S. 2018. State of the Art: Reproducibility in Artificial Intelligence. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*.

Kapoor, S.; and Narayanan, A. 2023. Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. *Patterns*, 4(9).

Kaufman, S.; Rosset, S.; Perlich, C.; and Stitelman, O. 2012. Leakage in Data Mining: Formulation, Detection, and Avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4).

Lin, Q.; Hsieh, K.; Dang, Y.; Zhang, H.; Sui, K.; Xu, Y.; Lou, J.-G.; Li, C.; Wu, Y.; Yao, R.; Chintalapati, M.; and Zhang, D. 2018. Predicting Node Failure in Cloud Service Systems. In *Proceedings of the 26th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*.

Marrone, M.; and Kolbe, L. M. 2011. Impact of IT Service Management Frameworks on the IT Organization: An Empirical Study on Benefits, Challenges and Processes. *Business & Information Systems Engineering*, 3(1): 5–18.

Notaro, P.; Cardoso, J.; and Gerndt, M. 2021. A Survey of AIOps Methods for Failure Management. *ACM Transactions on Intelligent Systems and Technology*, 12(6).

Remil, Y.; Bendimerad, A.; Mathonat, R.; and Kaytoue, M. 2024. AIOps Solutions for Incident Management: Technical Guidelines and A Comprehensive Literature Review. *arXiv preprint arXiv:2404.01363*.

Rendle, S.; Zhang, L.; and Koren, Y. 2019. On the Difficulty of Evaluating Baselines: A Study on Recommender Systems. *arXiv preprint arXiv:1905.01395*.

Steeman, W. 2013. BPI Challenge 2013, Incidents. 4TU.ResearchData. Volvo IT incident and problem management, VINST system.

Suriadi, S.; Andrews, R.; ter Hofstede, A. H. M.; and Wynn, M. T. 2017. Event Log Imperfection Patterns for Process Mining: Towards a Systematic Approach to Cleaning Event Logs. *Information Systems*, 64: 132–150.

Teinemaa, I.; Dumas, M.; La Rosa, M.; and Maggi, F. M. 2019. Outcome-Oriented Predictive Process Monitoring: Review and Benchmark. *ACM Transactions on Knowledge Discovery from Data*, 13(2): 17:1–17:57.

van der Aalst, W. M. P. 2016. *Process Mining: Data Science in Action*. Springer, 2nd edition.

van Dongen, B. F. 2014. BPI Challenge 2014. 4TU.ResearchData. Rabobank Group ICT, HP Service Manager; incident, change and interaction details.

Verenich, I.; Dumas, M.; La Rosa, M.; Maggi, F. M.; and Teinemaa, I. 2019. Survey and Cross-Benchmark Comparison of Remaining Time Prediction Methods in Business Process Monitoring. *ACM Transactions on Intelligent Systems and Technology*.

Weytjens, H.; and De Weerdt, J. 2022. Creating Unbiased Public Benchmark Datasets with Data Leakage Prevention for Predictive Process Monitoring. In *Business Process Management Workshops*, volume 436 of *Lecture Notes in Business Information Processing*, 18–29. Springer.

Williamson, B. D.; Gilbert, P. B.; Carone, M.; and Simon, N. 2021. Nonparametric Variable Importance Assessment Using Machine Learning Techniques. *Biometrics*, 77(1): 9–22.

Zhang, L.; Jia, T.; Jia, M.; Wu, Y.; Liu, A.; Yang, Y.; Wu, Z.; Hu, X.; Yu, P. S.; and Li, Y. 2024. A Survey of AIOps for Failure Management in the Era of Large Language Models. *arXiv preprint arXiv:2406.11213*.